



SEMINAR IN TOPICS OF INNOVATION I

**Importance Of ABM And Diffusion Innovation Process In Market Research.
Simulation Case Of Samsung And Apple In The Smartphone Market.**

Supervising Lecturer: Mathias Muller

Authors: Bianca Elena Costantin

Marta Molinari

Mirco Migliavacca

Academic Year: 2025

INDEX

1. INTRODUCTION	3
2. THEORETICAL BACKGROUND.....	4
2.1 What is Agent based modelling	4
2.2 Innovation diffusion process	4
2.2.1 Diffusion of innovation – s-curve	5
2.2.2 Word-of-Mouth diffusion process	7
3. MODEL ANALYSIS	9
3.1 Model Description	9
3.2 Results and Parameter experiments.....	9
3.3 Model Extension: Apple vs Samsung Competitive Setting	13
4. CONCLUSION	15
APPENDIX.....	18

FIGURE INDEX

Figure 1- Product Adoption Curve.....	7
Figure 2:- Initial State of the Model	10
Figure 3:- Final State of the model	10
Figure 4:- Graph about the Market Share	11
Figure 5:- Final State of the Modified Model	
Figure 6:- Graph about the Market Share of the Modified Model.....	12

TABLE INDEX

Table 1: - Characteristics of Adoption	5
Table 2:- Categories of Adopters.....	5

1. INTRODUCTION

This paper investigates the process of innovation diffusion driven by social interactions among consumers with heterogeneous characteristics.

Using an agent-based modelling approach, it shows how word-of-mouth amplifies the adoption rate of a new product, eventually leading to highly unbalanced adoption levels. It demonstrates that even a minimal variation in the initial adoption rate can be decisive for the acquisition of market shares. This is precisely the most interesting result of the analysis, that the success of an innovation is driven by social dynamics (Epstein, 1999).

To study this phenomenon, the model is inspired by consumers are heterogeneous and interact locally, individual adoption decisions act as a selection mechanism resulting in some innovations diffusing more rapidly, while others are excluded from the market.

The main contribution of this study is to show how the principles of Rogers (1983)' model of innovation diffusion can emerge from interactions among agents. In particular, the model reproduces and quantifies the S-shaped curve, which is generated because of social interactions and word-of-mouth.

This study has economic and managerial relevance. It allows an evaluation of how marketing strategies focused on consumers and early adopters can cause strong effects on final market shares. This result quantifies the impact of word-of-mouth and provides a useful tool for understanding how firms can build competitive advantages by exploiting the social mechanisms, instead of marketing strategies focused on intensive advertising (Zhang and Sun, 2020).

The paper is organized as follows, starting with the next chapter providing a review of the core literature and sets the stage for the model analysis carried out in Chapter Three. It shows the concepts of agent-based modelling and innovation diffusion, including reference to the Word-of-Mouth process. Chapter 3 presents the methodology and the results of the model and provides an economic and managerial interpretation.

2. THEORETICAL BACKGROUND

2.1 What is Agent based modelling

We are first going to define Agent-Based Modeling (ABM) which is the main concept of this paper. ABM can be viewed as a video game designed to solve all kinds of research problems. It's used to create a virtual world filled with individual agents which can represent people, companies, or even animals. You can give these agents simple rules, like copy what your friend buys or drive to work, and let them interact freely without a boss or other hard conditions controlling them. After starting the game, you can watch how their small, individual choices combine to create big, unexpected patterns, such as a traffic jam, a stock market crash, or a new technology going viral (Charles M. Macal & Michael J. North, 2014).

This is a great and really important simulation tool which can be used as a risk-free laboratory where you can safely test new ideas and what if scenarios without spending real resources. Unlike traditional ways where you can see just the final correlations and numbers, ABM reveals the actual mechanism, the actual cause and effect behind the success and or failure. - source

This study uses Agent-Based Modeling (ABM), which builds the market from the bottom up. As Epstein (1999) states, the core rule of this method is: “if you didn't grow it, you didn't explain it”. This implies that to truly explain a social pattern, such as why one brand dominates the market, we must show how it rises from the local interactions of independent agents (Epstein, 1999). In our specific model, we analyse how two dominating brands in the phone innovation market rise and compete against each other. Following the framework of Zhang and Sun (2020), we simulate these brands as a “heterogeneous population of agents” that compete in the same market segment. This approach allows us to observe the dynamic competition results and see which brand's innovation strategy eventually becomes the dominant strategy in the market.

ABM is crucial for this research because it can handle heterogeneity, which means that unlike older mathematical models that assume all agents are the same, ABM simulates individuals with different characteristics and states (Rand & Stummer, 2021). In our model, agents are differentiated by their status, like *potential adopters, innovators, or indifferent*. Consumer markets are described as “complex systems” (Rand & Stummer, 2021). In these systems, small random events, such as accidental social meetings, can lead to large, unexpected patterns that simple equations cannot predict (Rand & Stummer, 2021).

2.2 Innovation diffusion process

In the previous paragraphs we understood the importance of ABM and how this process allows us to predict the behaviour of agents. It can also be used to understand the diffusion process of an innovation within the market. In order to better understand this process, it is useful to clarify what is an innovation. According to Rogers (1983), one of the main experts of this phenomenon “An innovation is an idea, practice, or object that is perceived as new by an individual or other unit of adoption”.

Rogers(2003) wrote an important book, “Diffusion of Innovations”, in which he explains his theory about the diffusion process. There are five elements at the basis of the diffusion process: innovation, adopters,

communication channels, time, and social system (Rogers, 1983). Innovation is one of the most important aspects.

Regarding innovation, there are five specific characteristics that influence the speed of adoption (Bronwyn H. Hall, 2004).

Table 1: - Characteristics of Adoption

CHARACTERISTICS	EXPLANATION
Relative advantages	It is when an innovation is perceived as better compared to the idea it replaces. This concern economic profit, social status or other aspects.
Compatibility	It refers to when innovation is felt as coherent with values and needs of the adopters.
Complexity	It is the degree to which an innovation is perceived as complex in its use.
Trialability	It is when an innovation can be tried on a limited basis. Innovations that can be tested on a small scale are generally adopted more rapidly.
Observability	It is the degree to which an innovation is more observable than the others.

2.2.1 Diffusion of innovation – s-curve

The diffusion of innovation is “the process through which an innovation is communicated through certain channels, over time, among the members of a social system.” (Rogers, 2003).

According to the author, this process is based on four main elements: the innovation itself, communication channel, time and the social system (Rogers, 1983).

To understand how people decide to adopt an innovation, Rogers (1983) describes five-stage process.

Table 2:- Categories of Adopters

Type	Percentage	Description

Innovators	2,5%	These consumers are particularly willing to take risks, not fear to the complexity and uncertainty usually associated with new products
Early adopters	13,5%	They are not distrustful towards technology but need more information to better understand its functionalities, they are well integrated in their social system and have a high potential of influence on the behaviour of others;
Early majority	34%	They are more cautious and moves in a mass movement anticipating slightly the “average” market consumer;
Late majority	34%	People are sceptical towards innovation, because are more sensitive to price and waits for the uncertainty to decrease.
Laggards	16%	This people are those who understand little about technology and are averse to changes. Infact, they do not adopt a new product until the technology has become a standard and they have certainty of its usefulness.

The earliest adopters usually have higher socioeconomic status, higher education, greater open mindedness, better information capabilities and more social participation (Rogers, 1983).

Agents do not adopt simultaneously, but in sequential moments (Rogers,1983). An important role is played by the network of agents, within which people are reached by the innovation at intense levels (Rogers, 1983).

These categories can be used also for describe the diffusion process of an innovation (Rogers 1983). This process follows an S-shaped curve (sigmoid curve), from the introduction in the market to its diffusion (Rogers, 1983).

At the beginning adoption is slow, the curve is flat because only innovators take the risk (Rogers, 1983). Subsequently the innovation begins to be known and spread, in this way uncertainty is reduced, and the adoption rate grows quickly (Rogers, 1983). Finally, the market is saturated and only the laggards remain, the curve flattens again (Rogers, 1983).

As described by Kekäle and Helo (2014), the acceptance of a new product starts slow, but once it reaches a critical number of users, the sales grow exponentially. This critical number is a “Tipping Point”. According to Egbert et Al. (2016) it represents the moment when, over a certain threshold, the system moves on to a new state, in this case the mass adoption and widespread sales. This creates the S-shaped curve shown in the diagram.

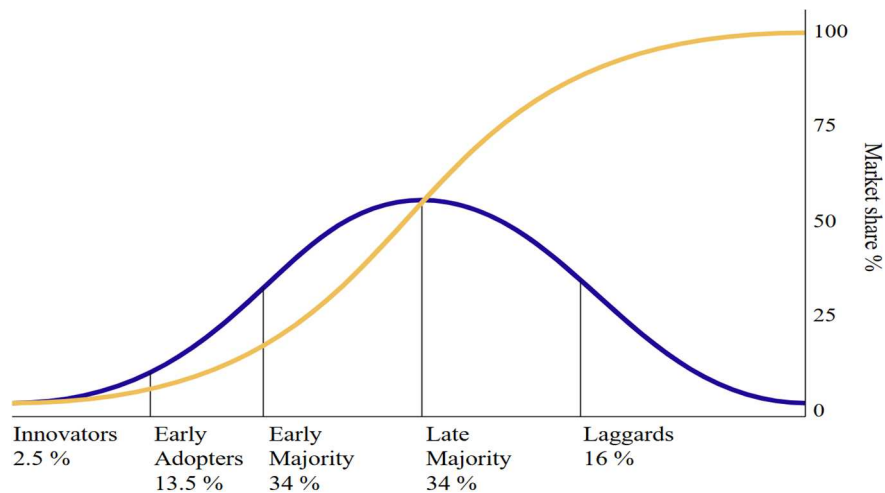


Figure 1- Product Adoption Curve Source: - Wikipedia By Rogers Everett - Based on Rogers, E. (1962) Diffusion of innovations. Free Press, London, NY, USA., Public Domain

This image illustrates the Product Adoption Curve (Dube & Gumbo, 2017). The graph presents two distinct lines representing (Dube & Gumbo, 2017). The explanation is found below:

Blue Line	The categories of adopters through a bell curve (normal distribution). It shows the rate of adoption over time.
Yellow Line	The Market Share. It is an S-curve. It shows cumulative adoption. It grows constantly until reaching 100% (market saturation).

The rate of adoption reflects the speed at which an innovation is adopted by the members of a social system (Rogers, 1983). It depends on several factors: the type of decision required, communication channels, characteristics of the social system and the commitment of change agents (Rogers, 1983).

2.2.2 Word-of-Mouth diffusion process

How many times have we met someone so excited about a new product that they feel the need to tell others how good it is? Or at the same time how many times we were negatively influenced by negative feedback? This process, known as Word-of-Mouth (WOM) communication, is considered one of the most powerful and fastest channels of communication and promotion in the market today (Leonard & Sensiper, 1998). Gildin (2002) in his seminal paper titled “Understanding the Power of Word-Of-Mouth” explains the main features of this process. Firstly, WOM relies on trusted personal sources such as friends and family and

not on-commercial sources. This factor translates directly into credibility as these communicators are perceived as free from any commercial interest. As a result, people consider their opinions more reliable, speeding up the product adoption processes. This consumer-driven power hides a dangerous duality. If on the one hand Positive WOM allows an intensive promotion channel, on the other hand negative WOM can be very damaging. As it had been demonstrated that negative experiences spread faster than positive ones (Gildin, 2002).

This dynamic is directly connected to the nature of innovation itself. Leonard and Sensiper (1998), in their seminal paper “The Role of Tacit Knowledge in Group Innovation”, explain the social process grounded in the exchange of tacit knowledge among individuals. They analyse that interactions are crucial for the diffusion of innovation in the market. In this sense WOM acts as a bridge between information sharing and market adoption of innovations. This is the crucial point, adoption comes because of personal experiences and judgments rooted in tacit knowledge. This makes WOM a decisive impulse in determining whether a new product succeed or fail.

3. MODEL ANALYSIS

This chapter introduces the Agent-Based Model used to simulate innovation diffusion and social interactions explored in the previous chapter, and the analysis of its dynamics inspired by the market competition between Apple and Samsung.

3.1 Model Description

In our simulation, the agents represent a “heterogeneous population” of consumers and firms. This means they are not all the same; they have different behaviors and states (pink, green, violet). We define these agents using Yoon's (2019) idea as "carriers" that contain specific genetic information. In our code, this “genetic information” is their color, which shows their specific "decision-making routines”. As Zhang and Sun (2020) explain, they are “bounded rational actors”. They do not know everything about the market. Instead, they possess the capacity to remember past experiences and can only find better strategies by learning, constantly adjusting and improving based on what they see around them.

The simulation happens in a "dynamic competitive market”. The agents are located on a grid where they move randomly and interact. According to Zhang and Sun (2020), this interaction happens in a "Moore neighborhood”. This means an agent is influenced by the neighbors immediately around it, on the same patch. This setup is important because it allows us to simulate local competition rather than just looking at the whole market at once.

The main rule in our model is based on "Genetic Information Transfer”. Yoon (2019) defines this as the moment when information is exchanged between the decision-making patterns of different agents. In our model, when an "Innovator" agent, in our case green or violet, meets a "Potential Adopter", in our case a pink agent, they transfer this information. The agents also follow a learning algorithm, after an interaction, they evaluate their success and make decisions for the next round by imitating the successful agents. This leads to a process of natural selection, where the agents with the best strategy survive and grow in numbers.

3.2 Results and Parameter experiments

It's important now to understand how our model works and what we can deduce from it referring to the topics discussed in the previous chapters. In this paragraph we analyse our model of diffusion.

In the beginning of our model, we have a population of agents representing potential adopters. These agents are all coloured pink because they have not yet come into contact with the innovation. Within this group we introduce a few different agents, the “early adopters” (Rogers, 2003). These are divided into two different colours, as shown in the *Figure 2*, Apple users purple and Samsung users green.

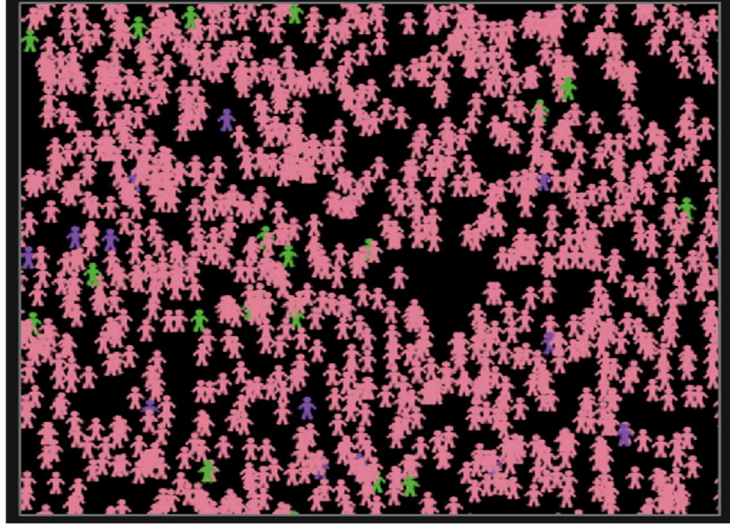


Figure 2:- Initial State of the Model

When we run the model, the agents move randomly and come into contact with their neighbours (Zhang and Sun, 2020). When two agents meet, there is a probability that one will be influenced by the other to adopt Apple or Samsung. During the entire diffusion process of the two innovations, we are able to observe how they divide the market.

However, not all the agents became green or purple, because the adoption is not guaranteed. An agent can turn into a third colour: red. These red agents are those who are not influenced by others, because they are indifferent and prefer a different smartphone brand (Figure 3).

Our model represented not a duopoly, but a scenario more realistic.

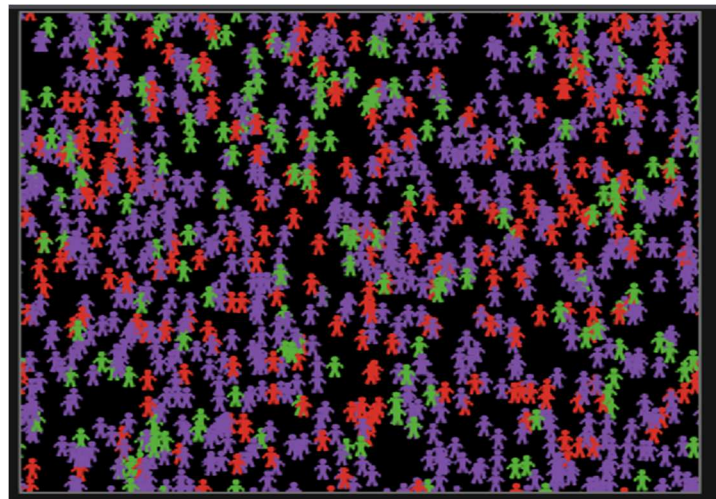


Figure 3:- Final State of the model

NetLogo is a programme used to understand the dynamics of this process through a graph. As shown in Figure 4, we are able to see how Apple and Samsung acquire new consumers and how their market share evolves over the time.

We can see in Figure 4 how the pink curve, which represents the agents at the beginning of the process, decreases until it reaches zero. The other curves have an S-shape curve and represent in purple Apple, in green Samsung and in red the neutral agents (Rogers, 2003).

We have used different parameters with our setting; Apple reaches a higher percentage of agents compared to Samsung. Meanwhile, the non-adopters (red) are more numerous than Samsung adopters.

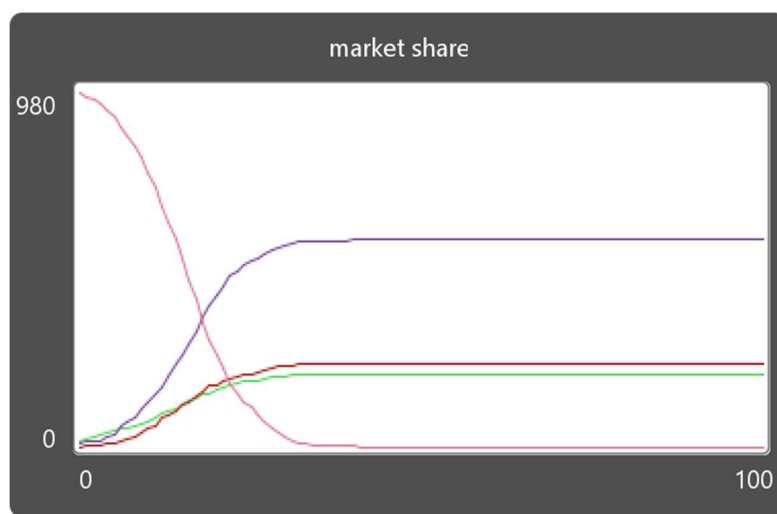


Figure 4:- Graph about the Market Share

As mentioned earlier, our model contains several parameters that can affect the result. First, we can change the number of agents in our model through the slider *number-turtles*. Combined with other parameters, this allows us to alter the speed of diffusion. We can also change the number of early adopters through two sliders: *n-apple* and *n-samsung*. This influences the result; for example, increasing *n-samsung* while reducing *n-apple*, will eventually lead Samsung to have the larger market share.

We obtain a similar result by modifying *percentage-apple* and *percentage-samsung*, which represent the probability that a neutral agent becomes infected by an Apple or Samsung adopter. The last slider in our model is *p* and it represents the proportion of potential adopters who are not influenced by others. These are the red agents who do not adopt Samsung or Apple.

This is an example of parameter modification through our slider:

Number-turtles: 100

n-apple: 13

n-samsung: 22

percentage-apple: 0.10

percentage-samsung: 0.20F

p: 0.8

The result of these modifications are shown in Figure 5 and 6.

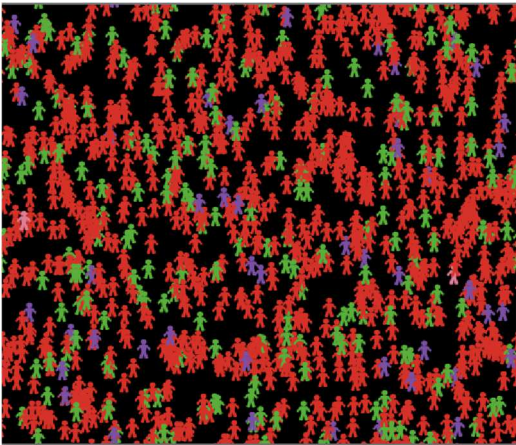


Figure 5:- Final State of the Modified Model

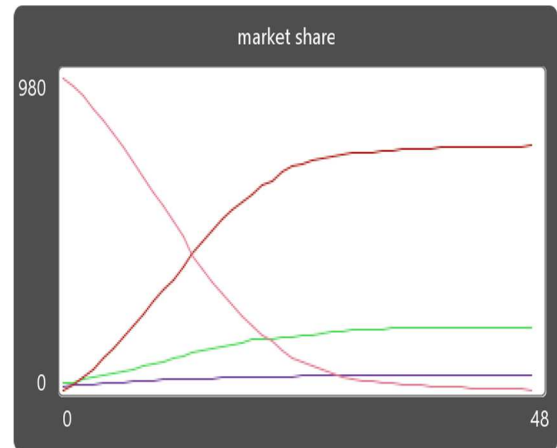


Figure 6:- Graph about the Market Share of the

Through this model we can better understand and observe some of the previously phenomena explained in Chapter 2 related to the diffusion of an innovation. Also, the whole code of the simulation can be found in the annex for a better understanding.

The graph in Figure 4 shows us three curves with an S-shape that perfectly represents what was explained in the previous chapter. We obtain Rogers' Curve (2003). The shape indicates that innovations at first grow slowly, after a tipping point accelerate, and then, when the market is saturated, flatten (Kekäle and Helo 2014). It is possible to see clearly the tipping point in the Graph of the figure 4.

In Chapter 2 we described how word-of-mouth is one of the main modes of innovation diffusion and its importance (Gildin, 2002). In our model, word-of-mouth is exactly the mechanism of influence, when two agents meet, one influences the other. This also highlights how diffusion is a social process.

3.3 Model Extension: Apple vs Samsung Competitive Setting

We chose to focus our ABM on the competition between Apple and Samsung because their duality shows a clear contrast in innovation and strategy. On the one hand, Apple's approach is based on a premium ecosystem and strong customer loyalty. On the other hand, Samsung's strategy is characterized by rapid innovation and a diversified product portfolio covering many price segments (Mayilsamy et al., 2025). Their differences translate directly into different patterns of customer choice.

Consequently, it would be interesting to analyse what would happen if we set different initial preferences for consumers before the adoption of the innovation. For example, older individuals might prefer some product characteristics instead of others that are preferred by younger people. In a complex model, heterogeneity in consumers' initial preferences is a relevant specification for simulating more realistic market scenarios.

Regarding the word-of-mouth diffusion, it would be interesting to set different levels of influence for agents. As shown by the research of Goldenberg, Libai and Muller (2001), the results of process are strongly connected to the "strong ties" and have closer communications into an individual's own personal group, also the "weak ties" form fewer personal communications among a wide set of acquaintances. In their analysis, the authors demonstrate that the accumulated influence of weak ties is even superior to the stronger ties (Goldenberg et Al., 2001). As a result, this extension would allow to model the interaction of these two ties and to verify if this pattern is confirmed also by this example.

All these extensions are useful for firms which, not by chance, are putting more and more focus on the final client in order to expand their market share.

3.4 Discussion

We found many findings, from which we consider the most interesting one the growing emergence of the S-shaped diffusion curve. Through our ABM model we can see how the pattern, shown in the example found in Figure 6, went higher naturally from micro-level interactions. This confirms Rogers' (2003) theory that social systems drive diffusion and validates Epstein's (1999) generative motto: "if you didn't grow it, you didn't explain it".

The initial parameters seen in the example also show that a small initial advantage in the number of early adopters, like slightly higher n_{samsung} , leads to market dominance in comparison with Apple. This aligns with David's (2000) concept of path dependence, where historical events lock the market into a specific trajectory, making it difficult for the losing brand to recover regardless of product quality.

Also, the most visible factor is the presence of red agents, who are indifferent, and act as a stopping factor of the diffusion. This factor is the reason why in our experiment the S-curve did not reach saturation,

being prevented by the red agents. This correlates with Kiesling et al. (2012), who argue that resistance leaders or negative word-of-mouth can significantly stall innovation diffusion. These agents break the chain of influence, acting as firewalls in the social network.

As a conclusion, our last finding was that increasing the number of agents and their interaction frequency accelerates the tipping point. This proves the findings of Zhang and Sun (2020) article, who noted that competition intensity is closely related to the evolutionary speed of innovation behavior. In a highly connected environment, the transition from early adoption to mass market happens much faster.

4. CONCLUSION

Through this study we wanted to demonstrate how the competitive dynamics are working between two major smartphone brands, Apple and Samsung, using an Agent-Based Modeling approach. We tried to create the market from scratch and analyze how individual consumer decisions, driven by set-up rationality and social interactions, lead to macroscopic market trends.

From a theoretical point of view our experiment confirms that the diffusion of innovation is an evolutionary process which is defined by important factors like “gradualism” followed by “episodic changes”. The model successfully simulated the transition where the market shifts from a fragmented state to a dominant design once a critical mass is reached. This shows the value of ABM in capturing emergent phenomena that static statistical methods often miss.

From a management point of view, the results suggest that companies should prioritize strategies that target early adopters and maximize word-of-mouth (WOM). As shown in our simulation, gaining a slight initial lead in the social network can result in a lock-in effect. Therefore, marketing strategies should focus not just on mass advertising, but on stimulating local interactions and influencing key opinion leaders to trigger the tipping point.

This study has a lot of limitations in comparison with the real world. The most important would be that the agents move randomly when real consumers are influenced by many more social, economical or other contextual factors which we couldn't reproduce. Future research could try to simulate more complex decision rules, such as price sensitivity or brand loyalty decay. Additionally, integrating “imperfect information” as suggested by Yoon (2019) could provide a more realistic representation of how consumers evaluate competing products in uncertain environments.

BIBLIOGRAPHY

1. C. Macal and M. North, "Introductory tutorial: Agent-based modeling and simulation," *Proceedings of the Winter Simulation Conference 2014*, Savannah, GA, USA, 2014, pp. 6-20, doi: 10.1109/WSC.2014.7019874. <https://ieeexplore.ieee.org/document/7019874>
2. David, Paul. (2000). Path Dependence, its critics, and the quest for 'historical economics'. https://www.researchgate.net/publication/23740891_Path_Dependence_its_critics_and_the_quest_for_historical_economics
3. Dube, C., & Gumbo, V. (2017). Diffusion of innovation and the technology adoption curve: Where are we? The Zimbabwean experience. *Business and Management Studies*, 3(3), 34-52. https://www.researchgate.net/profile/Victor-Gumbo/publication/317704759_Diffusion_of_Innovation_and_the_Technology_Adoption_Curve_Where_Are_We_The_Zimbabwean_Experience/links/595ae3fea6fdcc36b4d7bed0/Diffusion-of-Innovation-and-the-Technology-Adoption-Curve-Where-Are-We-The-Zimbabwean-Experience.pdf?sg%5B0%5D=started_experiment_milestone&origin=journalDetail
4. Egbert H. van Nes, Babak M.S. Arani, Arie Staal, Bregje van der Bolt, Bernardo M. Flores, Sebastian Bathiany, Marten Scheffer (2012). What Do You Mean, 'Tipping Point'?, *Trends in Ecology & Evolution*, Vol. 31, No 12pp 902-904. <https://doi.org/10.1016/j.tree.2016.09.011> .
5. Epstein, J. M. (1999). Agent-based computational models and generative social science. *Complexity*, 4(5), 41–60. [https://doi.org/10.1002/\(SICI\)1099-0526\(199905/06\)4:5%3C41::AID-CPLX9%3E3.0.CO;2-F](https://doi.org/10.1002/(SICI)1099-0526(199905/06)4:5%3C41::AID-CPLX9%3E3.0.CO;2-F)
6. Gildin, S. Z. (2002). Understanding the power of word-of-mouth. *Revista de Administração Mackenzie*, Vol 4(1), pp 91-106. [10.1590/1678-69712003/administracao.v4n1p92-106](https://doi.org/10.1590/1678-69712003/administracao.v4n1p92-106)
7. Goldenberg, Jacob & Libai, Barak & Muller, Eitan. (2001). Talk of the Network: A Complex Systems Look at the Underlying Process of Word-of-Mouth. *Marketing Letters*. Vol. 12. pp 211-223.
8. Kekäle, Tauno & Helo, Petri. (2014). The Tipping Points of Technology Development. *Quality Innovation Prosperity*. 18. 10.12776/qip.v18i1.211. https://www.researchgate.net/publication/274447381_The_Tipping_Points_of_Technology_Development
9. Kiesling, Elmar & Günther, Markus & Stummer, Christian & Wakolbinger, Lea. (2012). Agent-based simulation of innovation diffusion: A review. *Central European Journal of Operations Research*. 20. 183-230. 10.1007/s10100-011-0210-y. https://www.researchgate.net/publication/225831322_Agent-based_simulation_of_innovation_diffusion_A_review

10. Leonard, D., & Sensiper, S. (1998). The role of tacit knowledge in group innovation. *California management review*, 40(3), 112-132. https://www.researchgate.net/profile/Dorothy-Leonard-3/publication/230557658_The_Role_of_Tacit_Knowledge_in_Group_Innovation/links/5469fed90cf2f5eb180770de/The-Role-of-Tacit-Knowledge-in-Group-Innovation.pdf
11. Mayilsamy, C., Mithra, V., Vaisali, M., & Prannav, S. K. (2025). Comparative study of Samsung and Apple's mobile market performance and competitive insights. *YMER*, Vol. 24(3), pp.1027-1033.
12. Mayilsamy, C., Mithra, V., Vaisali, M., & Prannav, S. K. (2025). Comparative study of Samsung and Apple's mobile market performance and competitive insights. *YMER*, Vol. 24(3), pp.1027-1033. [10.54097/5q0mnb51](https://doi.org/10.54097/5q0mnb51)
13. Rand, William & Stummer, Christian. (2021). Agent-based modeling of new product market diffusion: an overview of strengths and criticisms. *Annals of Operations Research*. 305. 10.1007/s10479-021-03944-1. https://www.researchgate.net/publication/349186184_Agent-based_modeling_of_new_product_market_diffusion_an_overview_of_strengths_and_criticisms
14. ROGERS, E. M., (1983) Diffusion of Innovations. The Free Press of Glencoe, New York. <https://teddykw2.files.wordpress.com/2012/07/everett-m-rogers-diffusion-of-innovations.pdf>
15. ROGERS, E. M., (2003) Diffusion of Innovations. The Free Press of Glencoe, New York. https://books.google.com/books/about/Diffusion_of_Innovations_5th_Edition.html?hl=nl&id=9U1K5LjUOwEC
16. Yoon, J. (2019). *Study on the evolutionary patterns of product: agent-based simulation approach* (Doctoral dissertation, 서울대학교 대학원). <https://space.snu.ac.kr/bitstream/10371/162036/1/000000158123.pdf>
17. Zhang R, Sun B (2020). A competitive dynamics perspective on evolutionary game theory, agent-based modeling, and innovation in high-tech firms. *Management Decision*, Vol. 58 No. 5 pp. 948–966, doi: <https://doi.org/10.1108/MD-06-2018-0666>

APPENDIX

```
to setup
ca
create-turtles number-turtles [
  setxy random-xcor random-ycor
  set size 15
  set shape "person"
  set color pink
]
ask patches [set pcolor black]
ask n-of n-apple turtles with [color = pink] [
  set color violet
]
ask n-of n-samsung turtles with [color = pink] [
  set color green
]
reset-ticks
end
to go
ask turtles [
  fd 3
  rt random 360
]
ask turtles with [color = violet] [
  ask turtles-here with [color = pink] [
    if random-float 1 < percentage-apple[
      set color violet
    ]
  ]
]
ask turtles with [color = green] [
  ask turtles-here with [color = pink] [
    if random-float 1 < percentage-samsung [
```

```

    set color green
  ]
]
]

ask turtles with [color = violet] [
ask turtles-here with [color = pink] [
  if random-float 1 < p [
    set color red
  ]
]

ask turtles with [color = green] [
ask turtles-here with [color = pink] [
  if random-float 1 < p [
    set color red
  ]
]
]
]
tick
end

```